本课程包含大量复杂难懂的理论.
但也是有趣又有用的理论.

这里只整理应用部分. 学会怎么解题, 怎么考试.

1. Dynamic System. 分析动力系统. equilibrium / fixed point 的稳定性.

引入 Lyapunov 稳定性介绍.

Nonlinear.

Autonomous system

$\dot{\underline{x}} = \underline{f}(\underline{x})$. find equilibria $\underline{f}(\underline{x}^*) = 0$

linearize system near $\underline{x}^*$, Taylor series at $\underline{x}^*$

$$\underline{f}(\underline{x}) = \underline{f}(\underline{x}^*) + \underbrace{\left.\frac{\partial \underline{f}}{\partial \underline{x}}\right|_{\underline{x}=\underline{x}^*}}_{=: J(x^*)\ \text{Jacobian}} (\underline{x} - \underline{x}^*) + \text{H.O.T.}$$

✳ 得到 $J(x^*)$ 的 eigenvalue. λ_i.

① 所有 $Re(\lambda_i) < 0$: locally asymp. stable

② 存在 $Re(\lambda_i) > 0$: unstable

③ otherwise : no statement.

对于线性系统.: $Re(\lambda_i) \le 0$ stable

$Re(\lambda_i) < 0$ globally asy stable

$Re(\lambda_i) > 0$ unstable

存在且 $Re(\lambda_i) = 0$, unstable

每个 $\underline{x}^*$ 都可以这样判断, 一个系统常常存在多个 $\underline{x}^*$, 都可这样判断稳定性

这样的方法被称为 Lyapunov indirect method.

在这种方法下, 我们还为 equilibra 分了类

direction $\underline{v}_i$

stable subspace $E_s = \text{span}\{\underline{v}_i \mid Re(\lambda_i) < 0\}$

unstable subspace $E_u = \text{span}\{\underline{v}_i \mid Re(\lambda_i) > 0\}$

center subspace $E_c = \text{span}\{\underline{v}_i \mid Re(\lambda_i) = 0\}$

$\text{span}\{\underline{v}_i\} = \text{span}\{Re(\underline{v}_i), Im(\underline{v}_i)\}$

$x \in \mathbb{R}^2$

Node : stable node $\lambda_1 < 0$, $\lambda_2 < 0$ real

unstable node $\lambda_1 > 0$, $\lambda_2 > 0$ real

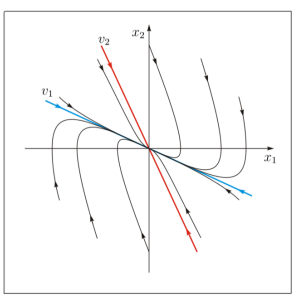


Fig. 2.1: Stable node

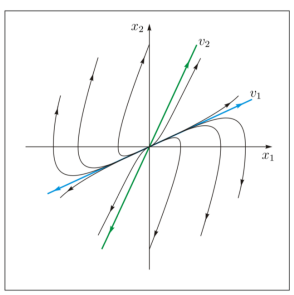


Fig. 2.2: Unstable node

Focus : $\lambda_1 = \bar{\lambda}_2$, $Im(\lambda_i) \neq 0$, $Re(\lambda_i) \neq 0$

stable focus $Re(\lambda_i) < 0$

unstable focus $Re(\lambda_i) > 0$

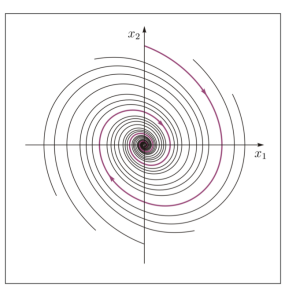


Fig. 2.3: Stable focus

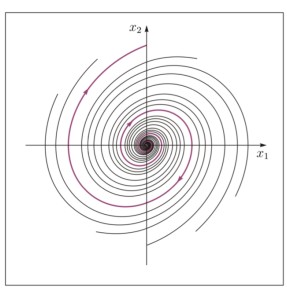


Fig. 2.4: Unstable focus

Center: $\lambda_1 = \bar{\lambda}_2$, $\mathrm{Re}(\lambda_i) = 0$, $\mathrm{Im}(\lambda_i) \neq 0$

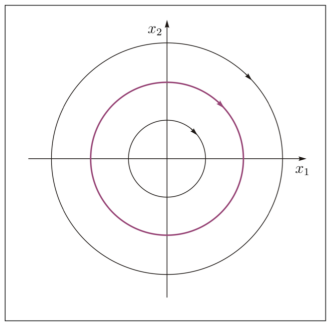


Fig. 2.5: Center

$E_c = \mathbb{R}^2$

Saddle : λ_i real , $\lambda_1 < 0$, $\lambda_2 > 0$ (or $\lambda_1 > 0$, $\lambda_2 < 0$)

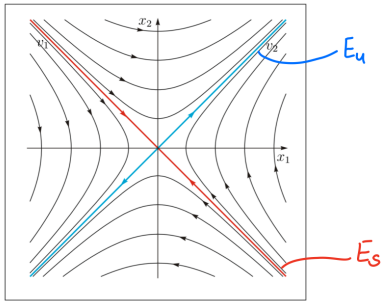


Fig. 2.6: Saddle

- 对于离散系统 $x_{i+1} = F(x_i)$, fixed point $x^* = F(x^*)$
- 同时，在这一章，我们还需要大概了解一下，periodic solution . 这两点在后面都会细讲.

for nonautonomous system $\dot{x}(t) = f(t, x)$

$\underline{x}^p(t) = \varphi(t, t_0, \underline{x}_0^p) = \varphi(t+T, t_0, \underline{x}_0^p)$ $\forall t$ $(t_0, \underline{x}_0^p)$ on solution curve

for autonomous system $\dot{x}(t) = f(x)$

$\underline{x}^p(t) = \varphi(t, t_a, \underline{x}_0^p) = \varphi(t+T, t_a, \underline{x}_0^p)$ $\forall t_a, t$ $\underline{x}_0^p$ on solution curve

$\Rightarrow f(t, x) = f(t+T, x)$

如果 $f(t, \underline{x}) = f(t+T, \underline{x})$, $kT = T$. period-k solution.

也被叫做 limit cycle

Quasiperiodic Solution : 比如 $x(t) = \cos \omega t + \sin \Omega t$. 不严格的循环.

- 再对 Chaotic 做一定的了解，它描述了一种 solution curve 相近又相离的情况.

① Sensitive dependent on intial conditions.

$\forall \delta > 0$, $\exists \varepsilon > 0$ for $\underline{x}_0 \in A$, $\exists \underline{y}_0 \in A$

s.t. $\|\underline{y}_0 - \underline{x}_0\| < \delta \Longrightarrow \|\underline{\varphi}(t, t_0, \underline{x}_0) - \underline{\varphi}(t, t_0, \underline{y}_0)\| > \varepsilon$

② topologically transitive.

$U \subset A$, $V \subset A$, $\underline{\varphi}(t, t_0, U) \cap V \neq \phi$

① + ② + periodic solutions dense in A , $\Longrightarrow$ chaotic.

(periodic-k orbits) for 离散 $x^* = F^k(x^*)$

- 介绍 Lyapunov function.

Lyapunov stable : $\forall \varepsilon > 0 \ \exists \delta > 0$, s.t. $\|\underline{x}_0 - \underline{x}^*\| < \delta \Rightarrow \|\varphi(t, t_0, \underline{x}_0) - \underline{x}^*\| < \varepsilon$

locally attractive : $\exists \delta > 0$ s.t. $\|\underline{x}_0 - \underline{x}^*\| < \delta \Rightarrow \lim_{t \to \infty} \|\varphi(t, t_0, \underline{x}_0) - x^*\| = 0$

globally if $\delta = +\infty$, i.e. $\forall \underline{x}_0 \in \mathbb{R}^n$

Lyapunov function $V(\underline{x})$: $V(\underline{0}) = 0$

- locally pos. def. (LPDF) if $\exists h > 0$, $V(\underline{x}) > 0 \quad \forall \underline{x} \in B_h \setminus \{0\}$

  $B_h := \{\underline{x} \mid \|\underline{x}\| < h\}$.
- pos. def. $V(\underline{x}) > 0 \quad \forall \underline{x} \in \mathbb{R}^n \setminus \{0\}$
- radially unbounded if $\|\underline{x}\| \to \infty \Rightarrow V(\underline{x}) \to \infty$

An equilibrium $\underline{x}^* = 0$ of $\dot{x} = f(x)$ is

- stable if $\exists V(\underline{x})$ LPDF, $h > 0$ and $\dot{V}(\underline{x}) \leq 0 \quad \forall x \in B_h$.
- locally asymp. stable if $\exists V(\underline{x})$ LPDF and $-\dot{V}(\underline{x})$ is PDF in $\forall x \in B_h$
- asymp. stable if $\exists V(\underline{x})$ PDF + radially unbounded and $-\dot{V}(\underline{x})$ PDF.

LaSalle

$V(\underline{x})$ PDF, s.t. $\dot{V}(\underline{x}) \leq 0$ for $\underline{x} \in B_h$ and $\dot{V}(\underline{x}) = 0$ only when $\underline{x} = 0$

$\underline{x}^* = 0$ locally asymp. stable

可以更精准的判断稳定性，但需要选择合适的 $V(\underline{x})$.

2. Bifurcations of Equilibria.

Bifurcations：我们先把它叫做分岔，指系统某些参数变化

引起了 EP, fixed Point, periodic solution 等，发生数量或状态变化。

① 列出运动方程. $\dot{\underline{x}} = \underline{f}(\underline{x})$

$\dot{x}_1 = x_2$

$\dot{x}_2 = x_1^2 + 2x_2$

或 $\ddot{x} = x^2 + 2\dot{x}$

② 计算 EP. $\underline{x}^*$

$\underline{f}(\underline{x}^*) = 0$ 或 $\dot{x}^* = 0, \ddot{x}^* = 0$ 解 x^*.

③ 运用 eigenvalue 或 Lyapunov 分析 EP 的稳定性.

$$J(\underline{x}^*) = \left.\frac{\partial \underline{f}(\underline{x})}{\partial \underline{x}}\right|_{\underline{x}=\underline{x}^*}$$

或 $\begin{cases} \ddot{x} = 2x + \dot{x} \\ \lambda^2 = 2 + \lambda \end{cases}$

$\ddot{x} = \underbrace{x^2}_{\text{需要先线性化}} + 2\dot{x}$

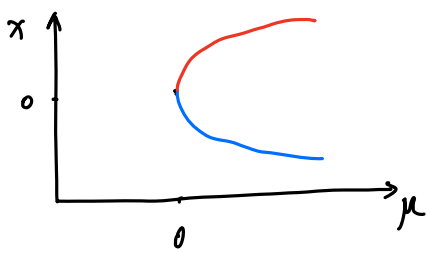
④ 根据结果画出 bifurcation diagram

给出一些 Normal Form.

- **Saddle - Node bifurcation** (turning point bifurcation)

$\dot{x} = \mu - x^2$

EP: $x_1^* = \sqrt{\mu}$ $x_2^* = -\sqrt{\mu}$ $\mu \geq 0$



$\frac{\partial f}{\partial x} = -2x$

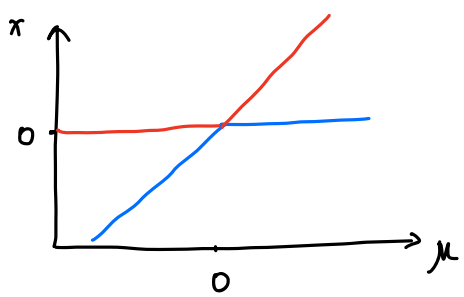
$\lambda_1 = -2\sqrt{\mu}$

$\lambda_2 = 2\sqrt{\mu}$

- **Transcritical bifurcation**

$\dot{x} = \mu x - x^2$

EP: $x_1^* = 0$ $x_2^* = \mu$



$\frac{\partial f}{\partial x} = \mu - 2x$

$\lambda_1 = \mu$

$\lambda_2 = -\mu$

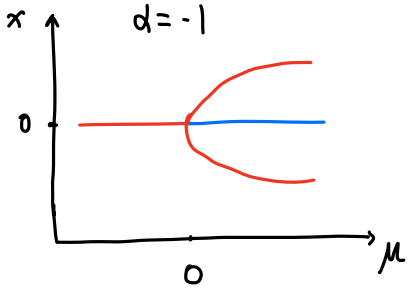
• Pitchfork bifurcation

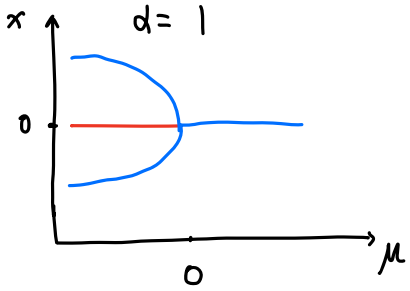
$\dot{x} = \mu x + \alpha x^3$, $\alpha = \pm 1$

EP: $x_1^* = 0$ $\quad x_{2,3}^* = \pm\sqrt{-\frac{\mu}{\alpha}}$ when $\frac{\mu}{\alpha} < 0$

$\frac{\partial f}{\partial x} = \mu + 3\alpha x^2$

$\lambda_1 = \mu \quad \lambda_2 = \lambda_3 = -2\mu$





• Hopf-bifurcation

$\dot{x} = \mu x - \omega y + (\alpha x - \beta y)(x^2+y^2)$

$\dot{y} = \omega x + \mu y + (\beta x + \alpha y)(x^2+y^2)$

$\underline{x} = \begin{bmatrix} x \\ y \end{bmatrix}$, EP: $\underline{x}^* = \underline{0}$ for $\|\underline{x}\|^2 < \left|\frac{\omega}{\beta}\right|$

$\left.\frac{\partial f}{\partial x}\right|_{\underline{x}^*} = \begin{pmatrix} \mu & -\omega \\ \omega & \mu \end{pmatrix}$, $(\mu-\lambda)^2 + \omega^2 = 0$, $\lambda_{1,2} = \mu \pm i\omega$

bifurcation 会在 $\mu = 0$ 时，也就是 $\lambda = \pm i\omega$ 时出现.

我们常使用极坐标转换.

$x = r\cos\theta$ $\quad \dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$

$y = r\sin\theta$ $\quad \dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

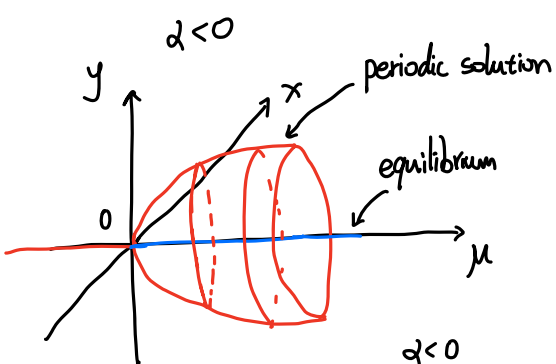
$\Longrightarrow$ $\dot{r} = \mu r + \alpha r^3$, $\dot{\theta} = \omega + \beta r^2$ $\quad r \neq 0$

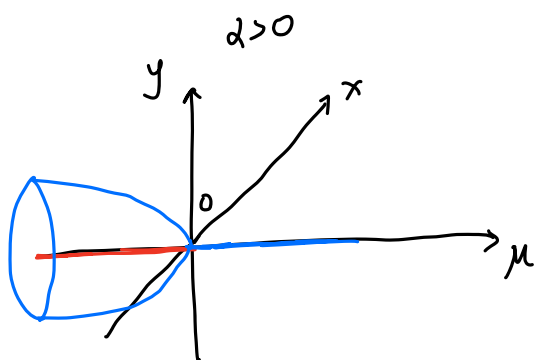
EP for $r \neq 0$.

$\dot{\theta} = \text{const} \neq 0$

$r^{*2} \neq -\frac{\omega}{\beta}$

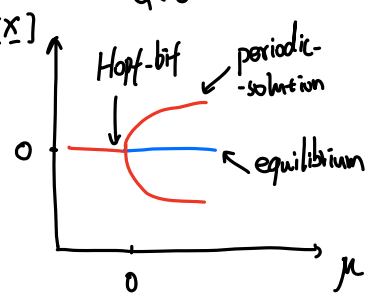
pitchfork bif. r 满足一个 pitchfork bif. $\dot{\theta} = \text{const.} \Longrightarrow$ periodic solution.



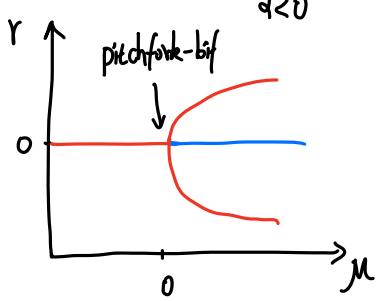


也可以这样表示

$[\underline{x}] = x^2+y^2$



因为它和



对应

本章有一个文件点，center manifold reduction.

$\Longrightarrow b$

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -a(x_2 + bx_1 - dx_1^2)$$

suspension trick

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -ax_2 - abx_1 + adx_1^2$$
$$\dot{b} = 0$$

center mainifold reduction. linearization around $\underline{x}^* = 0$

$$\begin{pmatrix} \underline{x} \\ b \end{pmatrix}^{\cdot} = \begin{pmatrix} A & 0 \\ \underline{0}^T & 0 \end{pmatrix} \begin{pmatrix} \underline{x} \\ b \end{pmatrix} + \begin{pmatrix} 0 \\ -ax_1 b + adx_1^2 \\ 0 \end{pmatrix}$$

这里 A 同等于 b=0 时的线性化.

$$A = \begin{pmatrix} 0 & 1 \\ 0 & -a \end{pmatrix} \quad \lambda_1 = -a \quad \underline{V}_1 = \begin{pmatrix} -1 \\ a \end{pmatrix} \quad E_s = \langle \underline{V}_1 \rangle$$
$$\lambda_2 = 0 \quad \underline{V}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad E_c = \langle \underline{V}_2 \rangle$$

Step 1: Transformation

这是有顺序的. 先 E_s, E_u, 再 E_c.

$$\begin{pmatrix} y \\ z \end{pmatrix} = T\underline{x}, \quad T^{-1} = (\underline{V}_1\ \underline{V}_2) = \begin{pmatrix} -1 & 1 \\ a & 0 \end{pmatrix}$$

$$\underline{x} = T^{-1} \begin{pmatrix} y \\ z \end{pmatrix} \Rightarrow \begin{matrix} x_1 = -y + z \\ x_2 = ay \end{matrix} \Longrightarrow \left.\begin{matrix} \dot{y} = -ay - b(z-y) + d(z-y)^2 \\ \dot{z} = -b(z-y) + d(z-y)^2 \\ \dot{b} = 0 \end{matrix}\right\}$$

EP点附近的线性化的 E_c、沿着 z 轴.

Step 2: Center manifold can locally be described by a function

$$y = \pi(z, b)$$
$$\Longrightarrow \dot{y} = \frac{\partial \pi}{\partial z}\dot{z} + \underbrace{\frac{\partial \pi}{\partial b}\dot{b}}_{=0} = \frac{\partial \pi}{\partial z}\dot{z}$$

Step 3: Taylor series expansion.

$$\pi(z, b) = c_1 z^2 + c_2 zb + c_3 b^2 + H.O.T.$$
$$\dot{y} = \frac{\partial \pi}{\partial z}\dot{z} = (2c_1 z + c_2 b)\dot{z}$$

substitute from $\dot{y}, \dot{z}$ in Step 1

$$\dot{y} = \dot{\pi} = -a\pi - b(z - \pi) + d(z - \pi)^2$$
$$(2c_1 z + c_2 b)(-b(z-\pi) + d(z-\pi)^2) = -a\pi - b(z-\pi) + d(z-\pi)^2$$

Step 4 : equate terms of order 2

把 π 代入.

z^2 : $0 = -ac_1 + d \Rightarrow c_1 = \frac{d}{a}$

bz : $0 = -ac_2 - 1 \Rightarrow c_2 = -\frac{1}{a}$

b^2 : $0 = -ac_3 \Rightarrow c_3 = 0$

Step 5 : determine dynamics on CM

$$\dot{z} = -b\left(z - \frac{d}{a}z^2 + \frac{b}{a}bz\right) + d\left(z - \frac{d}{a}z^2 + \frac{1}{a}bz\right)^2$$

$$\dot{z} = \underbrace{-b\left(1 + \frac{d}{a}\right)}_{=:\mu} z + \underbrace{d\left(3\frac{b}{a} + 1\right)}_{=:\alpha} z^2 + \text{H.O.T.}$$

$\dot{z} = -\mu z + \alpha z^2$ normal form of transcritical bifurcation.

3. Bifurcations of fixed Points

A fixed point $\underline{x}^*$.

$$x^* = F(x^*)$$

is locally asymp. stable if all eigenvalue λ_j of $\left.\frac{\partial F}{\partial \underline{x}}\right|_{\underline{x}^*}$ inside the unit circle.

$$|\lambda_j| < 1 .$$

对于线性系统：$|\lambda_j|<1$ globally asy. stable

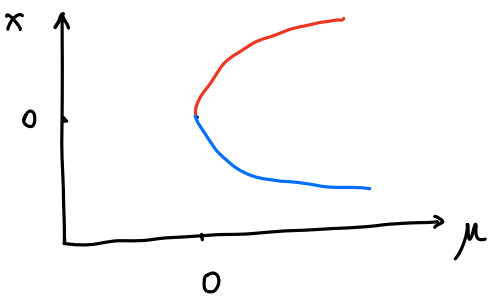
$|\lambda_{max}| = 1$ stable

- Turning point bifurcation.

$$x_{i+1} = x_i + \mu - x_i^2 = F(x_i)$$

fixed point : $x^* = x^* + \mu - x^*$, $x_1 = \sqrt{\mu}$ $x_2 = -\sqrt{\mu}$

$$\frac{\partial F}{\partial x_i} = 1 - 2x_i \qquad \lambda_1 = 1 - 2\sqrt{\mu} \qquad \lambda_2 = 1 + 2\sqrt{\mu}$$

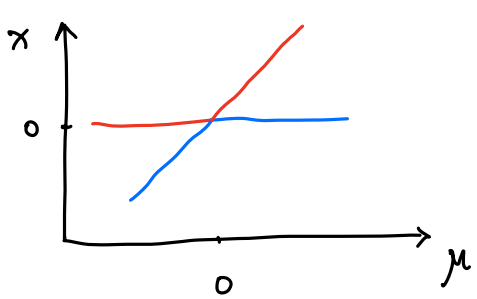


- Transcritical bifurcation

$$x_{i+1} = x_i + \mu x_i - x_i^2 = F(x_i)$$

fixed point : $x^* = x^* + \mu x^* - x^{*2}$ $x_1^* = 0$ $x_2^* = \mu$

$$\frac{\partial F}{\partial x_i} = 1 + \mu - 2x_i \qquad \lambda_1 = 1 + \mu \qquad \lambda_2 = 1 - \mu$$

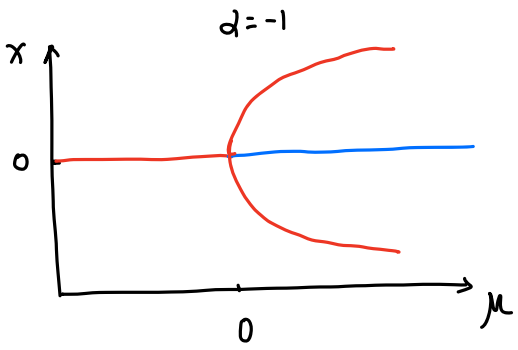


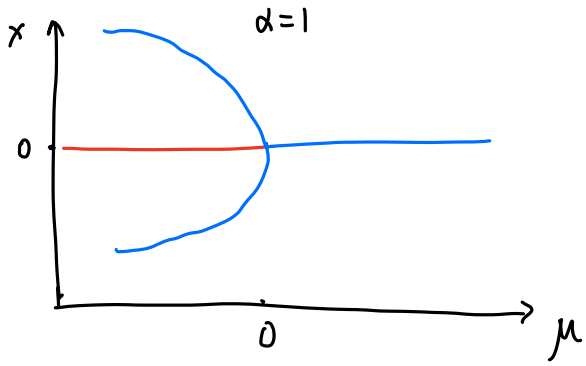
- Pitchfork bifurcation

$x_{i+1} = x_i + \mu x_i + \alpha x_i^3$

fixed point : $x^* = x^* + \mu x^* + \alpha x^{*3}$ $\quad x_1^* = 0$ $\quad x_{2,3}^* = \pm\sqrt{-\frac{\mu}{\alpha}}$ for $\frac{\mu}{\alpha} < 0$

$\frac{\partial F}{\partial x_i} = 1 + \mu + 3\alpha x_i^2$ $\quad \lambda_1 = 1+\mu$ $\quad \lambda_{2,3} = 1-2\mu$



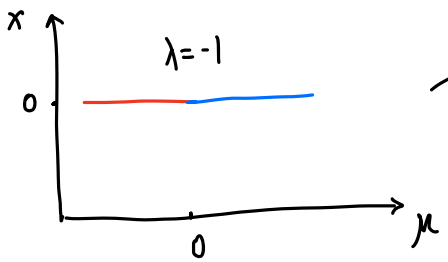


- Flip bifurcation

$x_{i+1} = -(1+\mu)x_i + x_i^3 = F(x_i)$

fixed point near origin $\quad x^* = 0$

$\frac{\partial F}{\partial x_i} = -(1+\mu) + 3x_i^2$ $\quad \lambda_1 = -(1+\mu)$



$\longrightarrow F^2(x_i)$

$x_1^{**} = 0$

$x_2^{**} = \sqrt{\mu}$

$x_3^{**} = -\sqrt{\mu}$

pitchfork bifurcation

- Niemark - Sacker bifurcation

$x_{i+1} = (\cos w\, x_i - \sin w\, y_i)(1+\mu+\alpha(x_i^2+y_i^2))$

$y_{i+1} = (\sin w\, x_i + \cos w\, y_i)(1+\mu+\alpha(x_i^2+y_i^2))$

Fixed point $(x^*, y^*) = (0,0)$

$A = (1+\mu)\begin{pmatrix} \cos w & -\sin w \\ \sin w & \cos w \end{pmatrix}$ $\quad \lambda_{1,2} = (1+\mu)e^{\pm jw}$

bif. 在 $\mu=0$ 时，也就是 $\lambda = e^{\pm jw}$ 时出现

$\Longrightarrow z_i = x_i + jy_i$. $\quad z_{i+1} = e^{jw} z_i (1+\mu+\alpha|z_i|^2)$

$z_i = r_i e^{j\theta_i}$ $\Longrightarrow r_{i+1} = r_i(1+\mu+\alpha r_i^2)$ $\longrightarrow$ pitchfork bifurcation .

4. Stability and Bifurcations of Periodic Solutions

- Autonomous system

$$\dot{\underline{x}}(t) = \underline{f}(\underline{x}(t)) \qquad \underline{x}(t_0) = \underline{x}_0$$

with solution $\underline{\varphi}(t, t_0, \underline{x}_0)$. has shift property.

$$\underline{\varphi}(t+\Delta t, t_0+\Delta t, \underline{x}_0) = \underline{\varphi}(t, t_0, \underline{x}_0) \quad \forall \Delta t.$$

他可能存在循环解.

$$\underline{\varphi}(t+T, t_0, \underline{x}_0^P) = \underline{\varphi}(t, t_0, \underline{x}_0^P) \quad \forall t.$$

- periodic Non-autonomous system

$$\dot{\underline{x}}(t) = \underline{f}(t, \underline{x}(t)) \qquad \underline{x}(t_0) = \underline{x}_0$$

with periodicity property of system.

$$\underline{f}(t, \underline{x}) = \underline{f}(t+T, \underline{x}) \qquad \forall \underline{x} \in \mathbb{R}^n, \forall t.$$

所以 has shift property

$$\underline{\varphi}(t+T, t_0+T, \underline{x}_0) = \underline{\varphi}(t, t_0, \underline{x}_0)$$

循环解:

$$\underline{\varphi}(t+T, t_0, \underline{x}_0^P) = \underline{\varphi}(t, t_0, \underline{x}_0^P) \qquad T = kT, \quad k \in \mathbb{N}^+$$

⇝ T shift: $\underline{\varphi}(t+T, t_0+T, \underline{x}_0) = \underline{\varphi}(t, t_0, \underline{x}_0)$

这两种系统在我们推判 periodic solution 稳定性时. 路有不同.

- Non-autonomous system.

$\underline{x}(t) = \underline{\varphi}(t, t_0, \underline{x}_0)$ 要先解出 $\underline{x}(t)$，再把 $\underline{x}_0$ 换成 $\underline{x}_i$，t 换成 t_0+T.

$\underline{f}(t,\underline{x}) = \underline{f}(t+T, \underline{x})$, $T = kT$.

$\Rightarrow \underline{P}_N(\underline{x}_i) = \underline{\varphi}(t_0+T, t_0, \underline{x}_i) = \underline{x}_{i+1}$

$\underline{P}_N : \mathbb{R}^n \to \mathbb{R}^n$ $\quad \underline{P}_N(\underline{x}_0) = \underline{\varphi}(t_0+T, t_0, \underline{x}_0) \Rightarrow \underline{x}(t_0+T) = \underline{P}_N(\underline{x}(t_0))$

fixed point $\underline{x}^* = \underline{P}_N(\underline{x}^*)$ is either a (i) periodic-k solution $\underline{\varphi}(t, t_0, \underline{x}^*) = \underline{\varphi}(t+T, t_0, \underline{x}^*)$

or an (ii) equilibrium $\underline{f}(t, \underline{x}^*) = 0$

$\left.\frac{\partial \underline{P}_N}{\partial \underline{x}}\right|_{\underline{x}^*} = \underline{\Phi}_T$ monodromy matrix.

eigenvalues of $\underline{\Phi}_T$ are λ_j. Floquet multipliers.

$|\lambda_j| < 1 \ \forall j$, then locally asy. stable.

$\underline{\Phi}_T$ 就是 $\underline{\Phi}(t)|_{t=T}$ 或 $\underline{\Phi}(t, t_0, \underline{x}^0)|_{t=t_0+T}$

- Autonomous system.

这里要用到一个复杂的 Poincaré map.

$\underline{x}(t) = \underline{\varphi}(t, t_0, \underline{x}_0)$

$\underline{x}_{i+1} = \underline{P}(\underline{x}_i) = \underline{\varphi}(t_0+T, t_0, \underline{x}_i)$

常用 $\underline{\varphi}(T, 0, \underline{x}_i)$

$\underline{P} : \mathbb{R}^{n-1} \to \mathbb{R}^{n-1}$ $\quad \underline{y}_{i+1} = \underline{P}(\underline{y}_i)$, $\quad \underline{y}_i = \underline{V}(\underline{x}_i)$

$\underline{V} : \mathbb{R}^n \supset \Omega \to \mathbb{R}^{n-1}$, $\underline{y} = \underline{V}(\underline{x})$ 降了一维

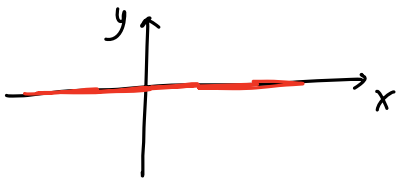
举例：$\Omega = \{(x,y) \in \mathbb{R}^2 \mid y = 0\}$, $\underline{n}_\Omega = [0,1]^T$ $\xrightarrow{\text{极坐标}}$ $y = r\sin\theta = 0$, $\theta = 2k\pi$

$\Omega = \{(r,\theta) \in \mathbb{R}^2 \mid \theta = 2k\pi, k \in \mathbb{N}, r > 0\}$

$\theta_i = 2\pi i$ $\quad x_i = r_i \underbrace{\cos\theta_i}_{=1} = r_i$

$r_{i+1} = P(r_i)$

$x_{i+1} = P(x_i)$



fixed point $\underline{y}^* = \underline{P}(\underline{y}^*)$ is either a (i) periodic solution $\underline{\varphi}(t, t_0, \underline{x}^*) = \underline{\varphi}(t+T, t_0, \underline{x}^*)$

$\underline{y}^* = \underline{V}(\underline{x}^*)$

or an (ii) equilibrium $\underline{f}(\underline{x}^*) = 0$

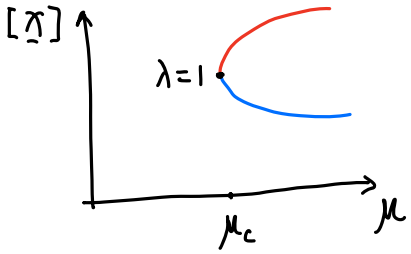
而且 eigenvalue 满足：

$\text{eig}(\underline{\Phi}_T) = \{1, \lambda_2, \cdots \lambda_n\}$

$\text{eig}\left(\left.\frac{\partial \underline{P}}{\partial \underline{y}}\right|_{\underline{y}^*}\right) = \{\lambda_2, \cdots \lambda_n\}$

$\text{eig}\left(\left.\frac{\partial \underline{P}}{\partial \underline{y}}\right|_{\underline{y}^*}\right)$ 满足 $|\lambda_j| < 1 \ \forall j$

then locally asy. stable.



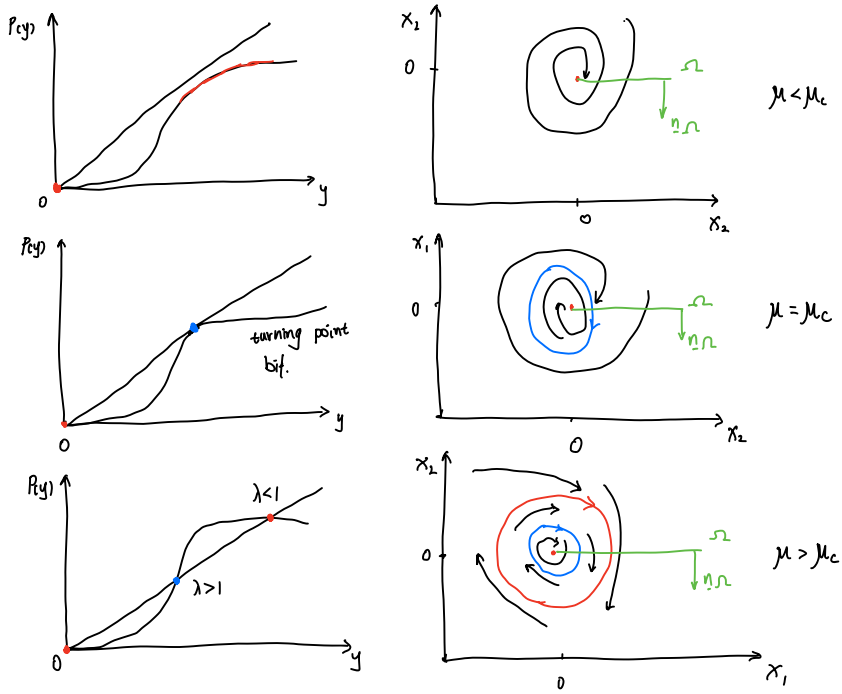
fold bifurcation.

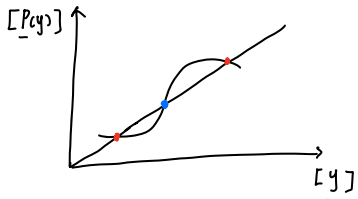
Symmetry-breaking bifurcation

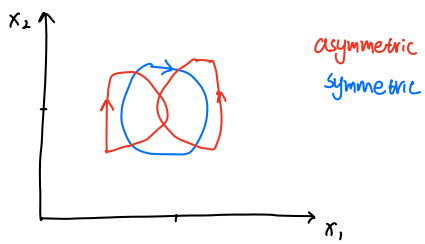
(only in non-autonomous)

$\underline{f}(t, \underline{x}) = f(t+T, \underline{x})$

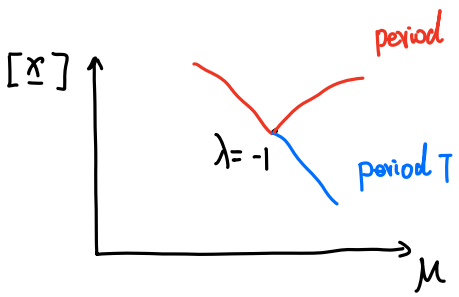
$\underline{f}(t, \underline{x}) = -f(t+\frac{1}{2}T, -\underline{x})$





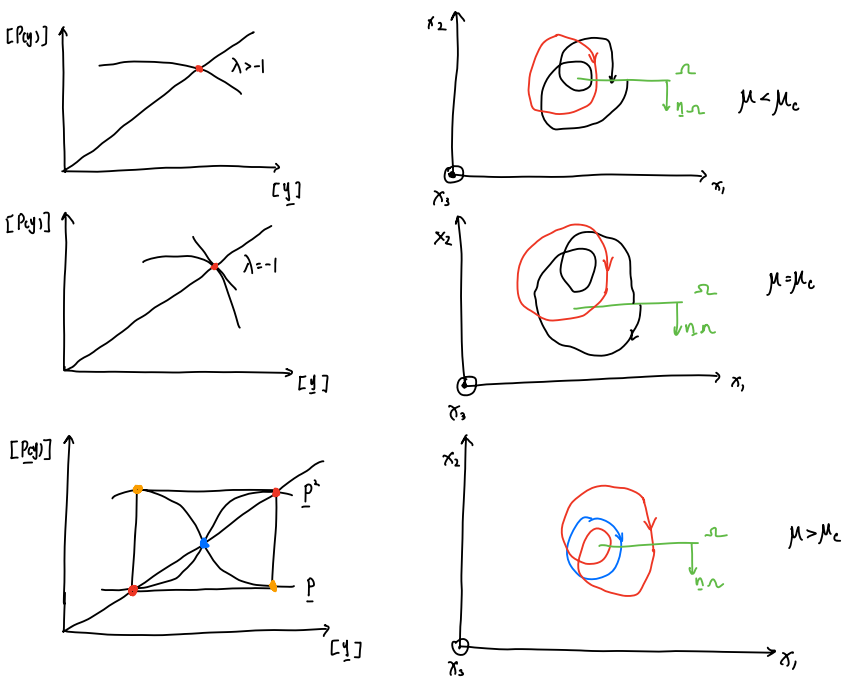


Floquet multiplier through +1



period-doubling bifurcation

(only for $n \geq 3$) or ($n \geq 2$ in non-autonomous)

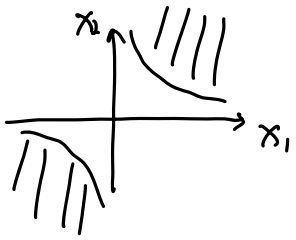


补充：

① 讨论 invariant set.

$T=\{\underline{x} \mid x_1 x_2 \geq 2\}$.

通过画 $x_2=\frac{2}{x_1}$ 曲线得到



如果 T 是 positive invariant set,

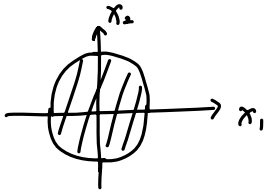
$\frac{d}{dt}(x_1 x_2) \geq 0, \forall x_1 x_2 \geq 2$

(初值选在这个 set，解也一直在这个 set 中)

那么 $\underline{x}^*=0$ 一定不是 GAS. 但可能是 LAS 或 stable 的.

因为选 x_0 在这个 set 中，也永远不会出这个集合，永远不会收敛到 $\underline{x}^*$.

$S=\{\underline{x} \mid x_1^2+x_2^2 \leq 2\}$



如果 S 是 positive invariant set.

$\frac{d}{dt}(x_1^2+x_2^2) \leq 0, \forall x_1^2+x_2^2 \leq 2$

无法说明 GAS. 如果 $\frac{d}{dt}(x_1^2+x_2^2)<0 \ \forall \underline{x} \in \mathbb{R}^2 \setminus \{0\}$.

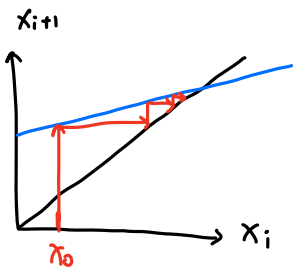
可以说明 LAS,

一定能说明 stable.

② $V(x_1,x_2)=\frac{1}{2}x_1^2+\frac{1}{4}(x_2+\mu(\frac{x_1^5}{5}-x_1))^2+\frac{1}{4}x_2^2$

$\dot{V}=x_1\dot{x}_1+\frac{1}{2}(x_2+\mu(\frac{x_1^5}{5}-x_1))(\dot{x}_2+\mu(x_1^4-1)\dot{x}_1)+\frac{1}{2}x_2\dot{x}_2$

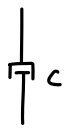
③ "cobweb"



④ Euler - Lagrange equations.

$L=T-V$.

$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}}-\frac{\partial L}{\partial q}=f_{NP}$



$f_{NP}=-c\dot{q}$.

在任何方向都取负

⑤ 求 $\Phi(t)$ (linear)

$\Rightarrow \dot{\underline{x}}=A\underline{x}+b(t)$

$\det(A-\lambda I) \longrightarrow \lambda_j$

遇复根 → $e^{iw}=\cos(w)+i\sin(w)$

$\lambda=\mu+iw$, $\underline{V}$. 此时 $\underline{V}$ 也可取复数.

$(A-\lambda_j I)\underline{V}_j=\underline{0} \longrightarrow \underline{V}_j$

$\Phi(t)=\left[e^{\lambda_j t}\underline{V}_j \ \cdots\right]$

$\Phi(t)=\left[\cdots e^{\lambda t}\underline{V} \ - \ - \ -\right]$

↓必须拆开写

$e^{\mu t}\begin{pmatrix}\cdot\\ \cdot\end{pmatrix}, e^{\mu t}\begin{pmatrix}\cdot\\ \cdot\end{pmatrix}$

Re Im

遇重根. $\lambda_m=\lambda_n=\lambda$

$(A-\lambda I)\underline{V}=\underline{0}$

$(A-\lambda I)\underline{W}=\underline{V}$

$\Phi(t)=\left[\cdots e^{\lambda t}\underline{V}, e^{\lambda t}\underline{W}+te^{\lambda t}\underline{V}, \cdots\right]$

$\underline{x}(t)=\Phi(t)\left(C+\int_0^t \Phi^{-1}(s)b(s)ds\right)$